

Multi-Modal Observations

Perception

Modeling

Planning

Execution

Actions

(a) Pipelined Approach

Multi-Modal Observations

Large Neural
Network



Actions

(b) End-to-end System

Multi-Modal Observations

Semantic
Modeling &
Planning

Visual
Prediction &
Planning

Reactive
Modeling &
Planning

Iterative
Sync

Execution

Actions

(c) CoStream System (Ours)